Kapil Wanaskar

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Research Engineer: Agentic AI, Post-training & Evals | 8 yrs building LLM agents, RL post-training, and benchmarks across Canva, AWS, and robotics.

Work Experience

Canva Research, *Senior Research Engineer (Agents & Evals)* Feb 2026 – Present | San Francisco, CA, USA
- Led long-horizon + agent trajectory evals and RL envs for 4+ teams on Canva Video Agent (CVA, 18k beta users)
- Trained CVA with agentic RL on a cost-vs-success reward, shortening trajectories to cut **p50 latency 45%**
- Built Pexels-Bench + a 330k video-mined edit-pair dataset that trained Canva Craft (#2 lab on DesignArena)

First American Title, *Senior ML Engineer* Apr 2025 – Feb 2026 | Remote, CA, USA
- Engineered MultiAgent workflow automating 250K+ tasks, cutting $1.25M cost at 98% accuracy [**Patent submitted**]
- Cut model costs 78% by distilling Qwen2.5-VL-7B to replace Gemini-2.5-Pro-API, 2.3x faster inference via Ollama

Amazon Web Services (AWS), *ML Engineer (Distributed Training)* May 2024 – Mar 2025 | Cupertino, CA, USA
- Implemented GRPO RL fine-tuning on Llama3.1, improving training efficiency **40% over PPO** across 100K+ samples
- Optimized distributed training on 100+ Trainium accelerators with int4 quantization, cutting memory use 60%
- Scaled multi-node distributed training with data + tensor parallelism and sharded optimizer state (ZeRO)

Intuitive Surgical, *ML Software Engineer* May 2023 – May 2024 | Sunnyvale, CA, USA
- Built FastAPI + VectorDB inference pipeline detecting out-of-distribution robotic-surgery logs at **98%** precision
- Post-trained embedding encoders and re-indexed FAISS via online RL, improving similarity-search security 13%
- Tuned 130K+ unsupervised-model variations on 150+ GB via SageMaker + MLflow, reaching 92% precision

Vectorr.in, *Software Engineer (Data Science)* Mar 2018 – Jul 2022 | Mumbai, India
- Engineered RFM + K-means customer segmentation on 10k+ daily visits, lifting satisfaction 3.1 to 4.8
- Built gradient-boosted uplift models for campaign targeting, amplifying marketing ROI **23%**

Research Publications

FactorJEPA on DENSEWORLD-115k: Can a World Model Predict Crowded, Chaotic South Asian Streets? 2026
- **World-Model** FactorJEPA adapts V-JEPA 2.1 through a 3-stage layout, agent and interaction training curriculum
- After 115k-clip training, separated from the strongest rival on all 4 predictive diagnostics (13.9–43.3 CI widths)
- Released DENSEWORLD-115k: 115,687 clips (276 h) from 714 YouTube videos across 22 Indian cities

SwiPO on ECLIPTICA-30k: Can You Trust One LLM Checkpoint to Switch Its Alignment for Every Agent Role? 2026
- Reached **86.7%** relative instruction-alignment efficiency on Llama-3.1-8B, leading GRPO 36.1% and PPO 20.4%
- Built ECLIPTICA-30k: 300 prompts × 100 alignment-instruction types, evaluated on the 3,000-case ECLIPTICA-3k
- Developed SwiPO (Switchable Preference Optimization): instruction-conditioned DPO under a mandatory KL anchor

A Comprehensive Dataset for Human vs. GenAI Image Detection 2026
- Created MS COCOAI dataset with 96,000 semantically-aligned real/synthetic pairs from 5 text-to-image diffusion models
- Established dual-task benchmark achieving **80.1%** binary detection and 44.9% model attribution
- Designed frequency-domain ResNet-50 robustness framework, detecting with 80.1% accuracy across 4 perturbations

Multimodal Benchmarking and Recommendation of Text-to-Image Generation Models, *IEEE CISOSE 2025* 2025
- Received the "BDS **Best Student Paper**" award
- Designed Weighted Score metric combining CLIP-Score, LPIPS, FID, MRR& Recall@3 via min–max normalization
- Metadata-augmented models (Flux, InContext LoRA) showed ~19% higher Weighted Score & ~15 point FID reduction

Prompt Recommendations for AI art, *IEEE AIKE, California, USA* 2023
- Extracted features of 5000 images via text embeddings and ensemble models
- Proposed graph-based evaluation of 3 recommendation and community-detection algorithms lacking rating signals

Education

MS in Artificial Intelligence, Computer Engineering, CA, USA
San José State University

Master of Computer Integrated Manufacturing and Bachelor of Engineering, *Indian Institute of Technology (IIT) Bombay* Mumbai, India